

Chapter 4

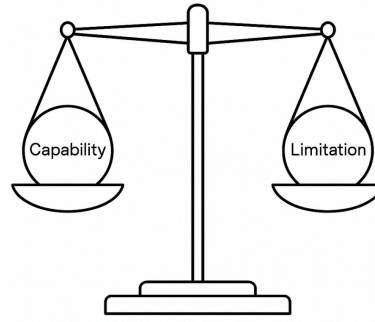
LLM Limitations: Hallucinations, Context Length, and Reasoning

Module 1: Foundations of Large Language Models

Every powerful system has boundaries. In the last three chapters we covered what LLMs are, how they're built, and how they process language. Now we're going to look at where they break down — and why that matters for anyone building with them.

Why Limitations Matter

Every capability has its limits



LLMs look intelligent. They write fluently, answer questions, generate code. But they don't understand the way humans do. They generate patterns that are statistically consistent with their training data — they're not reasoning, they're predicting. Knowing where that breaks down is how you use them responsibly.

What is a Hallucination?

Plausible but false



A hallucination is when a model produces something plausible but completely wrong. It's not lying — it has no concept of truth or falsehood. It's doing what it always does: predicting words that sound right given the context. The problem is that "sounds right" and "is correct" are not the same thing.

Why Hallucinations Happen



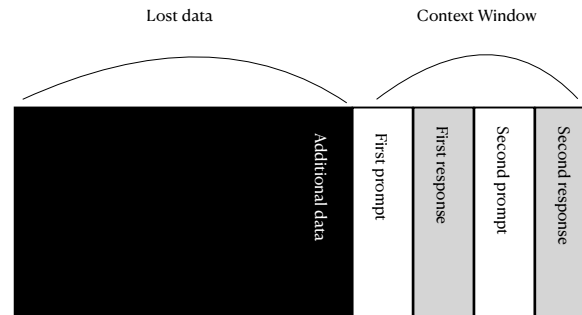
The model has no live connection to the world. Everything it knows comes from training data with a cutoff date. So when you ask about something obscure, recent, or outside its training — it doesn't say "I don't know." It fills the gap. Confidently. That's the dangerous part.

Spotting Hallucinations

Symptom	Example	Mitigation
Overconfidence	"The Eiffel Tower was built in 1950."	Add references / retrieval grounding
Fabrication	"According to the *Journal of Quantum Gardening*..."	Use verified datasets
Drift	"FusionAuth supports OIDC 3.0"	Use system prompts to constrain scope

There are three patterns worth recognizing. Overconfidence — a specific fact stated with total certainty that turns out to be wrong. Fabrication — a made-up source cited as real, like a journal that doesn't exist. And drift — where the model slides into territory it wasn't scoped for and starts inventing details. The mitigation in every case is grounding: give the model real data to work from, and verify anything high-stakes.

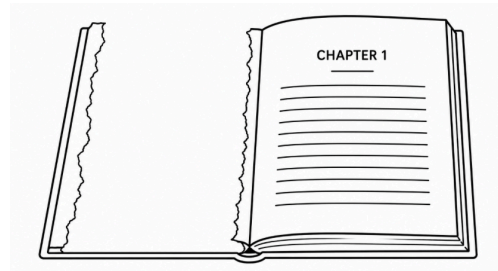
Context Length Explained



The context window is the model's working memory — everything it can see at once. Your prompt, the conversation history, any documents you've passed in — all of it counts against that limit. Older models topped out around 4,000 tokens. Many current models handle hundreds of thousands. But the rule holds: if it's outside the window, the model has no idea it ever existed.

Lost Context

Repetition or forgetting



When a conversation runs long enough that earlier content falls out of the window, the model loses access to it entirely. That's why you'll sometimes see a model contradict something it said twenty messages ago — it literally can't see that far back anymore. And even when you're well within the window, cramming in too much information at once can dilute the model's focus and produce vaguer, less useful responses.

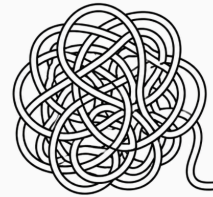
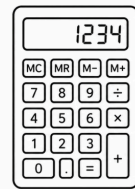
Reasoning and its Limits

Pattern

Logical

Recognition

Deduction



The good news is reasoning can be scaffolded. Chain-of-thought prompting — breaking a problem into explicit steps — meaningfully improves accuracy. Giving the model access to tools like calculators removes the need to "do math" at all. And RAG, which we'll spend a lot of time on later in this course, grounds responses in verified external data. The model reasons better when you structure the problem for it.

Improving Reasoning Performance

All together now



Three limitations worth keeping in your head whenever you work with LLMs. Hallucinations happen when the model lacks grounding. Context length defines what the model can actually see and remember. And reasoning is simulated — fluent and convincing, but not foolproof. Understanding these isn't about being cynical about the technology. It's about using it well. And building systems that account for where it fails is exactly what the rest of this course is about.

Recap and Reflection



- Know the limits
- Use wisely
- Design with care

Let's recap:

Hallucinations show up when a model lacks grounding.

Context length defines what the model can remember.

Reasoning is simulated, not human-style logic.

As LLM practitioners, our power comes not from blind trust — but from clear understanding.

Knowing these limits is how we use LLMs responsibly, creatively, and effectively. Thanks for your attention for this background information - now that we have an idea what these LLMs are doing, let's get hands-on with prompt engineering - coming up in the next module!